CRYPTO TRADING PERFORMANCE ANALYSIS

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A Complete End-to-End Study

From Raw On-Chain Data to a Deployable Predictive Model

211,224	**246**	**32**	**2023 – 2025**
Fill Executions	Coins Traded	Accounts	Date Range

1. Project Overview

This project analyses real on-chain cryptocurrency trading data to answer two questions: does market sentiment predict trade profitability, and can a machine-learning model reliably identify which trades are worth taking?

Starting point: 211,224 raw trade executions from a live crypto account, spanning May 2023 to May 2025.

Ending point: An XGBoost model achieving ROC-AUC = 0.91 on genuine out-of-sample future data, with a 61% win rate at the recommended decision threshold.

Key discovery: The first naive model showed AUC = 0.99. This was entirely fake — data leakage. Fixing it required restructuring the entire dataset from scratch.

The Four Stages

Stage	What We Did	Key Result
1 — Data Prep & EDA	Load, clean, merge with Fear & Greed Index, explore distributions and sentiment patterns	Confirmed sentiment affects PnL; Extreme Greed = best win rate (46.5%)
2 — Baseline Model	Random Forest on fill-level data with a random train/test split	AUC 0.99 — entirely false, caused by three compounding leakage errors
3 — Leakage Fix	Remove leaky features; switch to a temporal (chronological) split	AUC collapses to 0.47 — correctly reflects a real market regime break
4 — Final Model	Reconstruct trade-level data; engineer momentum, volatility, rolling win-rate features; retrain XGBoost	AUC 0.91 on future test data — genuine, deployable predictive power

2. Understanding the Data

2.1 What Is a Fill?

When a trader places a large order — say, buy $50,000 worth of Bitcoin — the exchange rarely executes it as one transaction. Instead, the order book fills it piece by piece across multiple price levels as liquidity becomes available. Each of these sub-executions is called a fill.

> Example: A buy order for 1,000 ETH might generate 30 separate fills — some at $3,000, some at $3,001, some at $3,002 — depending on how much liquidity exists at each price level. The dataset contains all 30 of those fills as individual rows, not as one trade.

This distinction — fill vs trade — turned out to be one of the most important discoveries in the entire project. Treating fills as independent trading decisions introduced a structural artifact that made the first model's results completely misleading.

2.2 Fill-Level Dataset Schema

Column	Type	Meaning & Why It Matters
Account	Text	On-chain wallet address. 32 unique accounts in the dataset.
Coin	Text	Asset traded: BTC, ETH, SOL, HYPE, FARTCOIN, etc. 246 unique coins.
Execution Price	Numeric	Exact price this fill was matched at. Varies across fills of the same order.
Size Tokens	Numeric	Tokens bought/sold in this fill. Mathematically = Size USD / Execution Price.
Size USD	Numeric	Dollar notional value of this fill.
Side	BUY / SELL	BUY fills open a position. SELL fills close it and realise PnL.
Timestamp IST	DateTime	Exact execution time in India Standard Time (UTC+5:30).
Closed PnL	Numeric	Profit or loss realised at this fill. Always zero for BUY fills.
Order ID	Integer	Groups all fills belonging to the same order. The key for trade-level aggregation.
Fee	Numeric	Exchange fee paid. Proportional to Size USD.
Start Position	Numeric	Cumulative position size after this fill. Path-dependent — leakage risk.

Critical: Closed PnL = 0 for every BUY fill. Only SELL fills realise a profit or loss. This is not noise — it is how leveraged trading works. Any model trained on fill-level data will trivially learn that SELL = possibly profitable and BUY = never profitable. That is a circular structural artifact, not a tradeable signal.

2.3 The Fear & Greed Index

The Fear & Greed Index is a daily composite sentiment indicator for the crypto market. It runs from 0 (Extreme Fear) to 100 (Extreme Greed) and is computed from six sub-indicators:

- **Volatility (25%):** Current volatility versus 30-day and 90-day averages. Unusually high volatility signals fear.
- **Market Momentum / Volume (25%):** Current volume versus 30 and 90-day averages. Buying on high volume signals greed.
- **Social Media (15%):** Hashtag interaction rates on crypto-related content.
- **Surveys (15%):** Weekly polling of crypto market participants.
- **Bitcoin Dominance (10%):** When BTC's market share rises, money is rotating out of altcoins — a fear signal.
- **Google Trends (10%):** Search volume for terms like 'Bitcoin' and 'crypto crash'.

Score	Label	Market Interpretation
0 – 24	Extreme Fear	Panic selling; historically a contrarian buy signal
25 – 44	Fear	Bearish sentiment dominates; investors cautious and defensive
45 – 55	Neutral	No strong directional conviction in the market
56 – 74	Greed	FOMO (Fear Of Missing Out) beginning to drive buying
75 – 100	Extreme Greed	Market overheated; elevated risk of correction

3. Data Preparation & Merging

3.1 Parsing and Cleaning

The raw CSV contained timestamps in the non-standard format DD-MM-YYYY HH:MM, requiring explicit parsing configuration. Several numeric columns arrived as strings with artefacts that needed coercion. After cleaning and validation: 211,218 fills retained across a date range of 2023-05-01 to 2025-05-01.

3.2 The Merge Strategy

The two datasets share no direct key. Fills are timestamped to the minute; Fear & Greed values are published once per calendar day. The merge extracts the calendar date from each fill timestamp and performs a left join on that date. Six rows with no matching sentiment date were dropped (less than 0.003% of data).

Columns added after merge:

fg_value — numeric score (0–100) for that trading day

fg_class — categorical label (Extreme Fear, Fear, Neutral, Greed, Extreme Greed)

hour — hour of execution (0–23); crypto markets run 24/7 but liquidity varies by session

day_of_week — 0 = Monday through 6 = Sunday; weekend trading has different volatility characteristics

is_profitable — binary target: 1 if Closed PnL > 0, else 0

4. Exploratory Data Analysis (EDA)

EDA is the process of understanding the data before building models. It reveals distributions, relationships, anomalies, and the stories hidden in summary statistics. Every modelling decision downstream was informed by what we found here.

4.1 Distribution Analysis

Both Size USD and Closed PnL are strongly right-skewed — the classic shape of financial data. A small number of very large fills dominate the total volume, while the majority are small.

Size USD

- Median: ~$597 (half of all fills are below this)
- Mean: significantly higher due to large outliers
- Long right tail: a handful of fills represent a disproportionate share of total volume

Closed PnL

- Median: $0.00 — most fills are BUY fills with no realised PnL
- Positive tail: extends to +$600,000 for very large winning closes
- Negative tail: extends to -$400,000 for large losing closes
- The extreme range reflects leveraged position trading

4.2 Correlation Analysis

A correlation matrix measures the strength and direction of linear relationships between numeric variables. The key values found:

Variable Pair	Correlation	Interpretation
Fee vs Size USD	0.75	Strong positive — fees are charged proportionally to notional value
Size Tokens vs Size USD	0.68	Expected — they are linked mathematically through Execution Price
fg_value vs Closed PnL	0.01	Near-zero linear relationship — but sentiment matters non-linearly (tree models find this)
Execution Price vs Closed PnL	0.04	Weak — the price level at execution does not determine outcome

> Important: A near-zero linear correlation does not mean a feature is useless. Tree-based models like XGBoost discover non-linear and conditional relationships that linear correlation completely misses. Sentiment at a value of 80 means something very different from sentiment at 20 when combined with a strong price momentum signal.

4.3 Sentiment Performance Analysis

Grouping trades by their daily sentiment classification revealed the most important pattern in the dataset — and a result that contradicts conventional trading wisdom:

Sentiment	Avg PnL/Fill	Win Rate	Volume Traded	Fill Count
Extreme Fear	$34.54	37.1%	$114M	21,400
Fear	$54.29	42.1%	$483M	61,837
Neutral	$34.31	39.7%	$180M	37,686
Greed	$42.74	38.5%	$289M	50,303
Extreme Greed	$67.89	46.5%	$124M	39,992

- **Extreme Greed = best performance:** Highest average PnL ($67.89) and highest win rate (46.5%). This contradicts the 'buy fear, sell greed' contrarian wisdom, suggesting the trader uses a momentum strategy — riding trends that are strongest during Greed periods.
- **Fear = highest volume:** $483M traded during Fear periods, the most of any category. The trader is most active when markets are moderately bearish — likely accumulating positions at lower prices.

Statistical validation: A Kruskal-Wallis test confirmed that PnL distributions differ significantly across sentiment groups ($p < 0.05$). This is a non-parametric test appropriate for non-normal, skewed financial data. It asks whether any group has a different distribution without assuming bell-curve shapes.

4.4 Time-of-Day Patterns

Hour 12 IST (12:00 noon India time = 06:30 UTC, the London pre-market open) showed the highest average PnL at approximately $130 per fill. Hour 23 showed the lowest at approximately $18. These patterns reflect liquidity cycles across the Asian, European, and US trading sessions — a well-documented phenomenon in 24/7 markets.

5. Baseline Model — The Leakage Trap

The first model was built using the most common approach — and produced a spectacular-looking result that was entirely wrong. Understanding exactly why it was wrong is the most educational part of the entire project.

5.1 What We Built

A Random Forest Classifier trained on six fill-level features: Execution Price, Size USD, Side (encoded as BUY=0 / SELL=1), fg_value (Fear & Greed score), hour, and day_of_week. Target: is_profitable. The data was shuffled randomly and split 80% training / 20% test — standard practice for non-time-series problems like image classification.

> Result: ROC-AUC = 0.9922. The model appeared to predict trade profitability with 99.2% discriminative power. This number is completely false.

5.2 What is ROC-AUC?

The ROC (Receiver Operating Characteristic) curve plots the True Positive Rate (fraction of profitable trades correctly identified) against the False Positive Rate (fraction of unprofitable trades wrongly flagged as profitable) across every possible decision threshold. The Area Under this Curve (AUC) summarises model quality in a single number:

- AUC = 0.5 — no better than random coin-flipping
- AUC = 0.70 — weak but potentially useful signal
- AUC = 0.85 — strong discriminative power
- AUC = 1.0 — perfect prediction
- AUC < 0.5 — model predictions are systematically inverted (worse than random)

5.3 Three Compounding Leakage Errors

Error 1: Random Split on Time-Series Data (Most Damaging)

A random split shuffles rows before dividing training from test sets. For cross-sectional data (images, customers, molecules), this is correct. For financial time-series data, it creates catastrophic look-ahead bias.

When rows are shuffled, a trade from April 2025 lands in the training set while a trade from June 2024 ends up in the test set. The model sees the future during training — it learns from tomorrow's data to predict yesterday's outcomes. In a rising market, this makes recent profitable trades bleed backward into training data for earlier periods, creating completely artificial predictive signal.

Error 2: Execution Price as a Time Proxy

Execution Price was the single most important feature in the baseline model at 51.5% importance. The reason reveals the leakage: in a bull market (which this dataset covers in 2024), prices trend monotonically upward. A high execution price therefore means the trade happened recently. Recent trades were more profitable because the overall market was rising. The model learned time trends, not trading signals — a completely non-causal relationship.

Error 3: Mathematical Feature Redundancy

Because **Size Tokens = Size USD / Execution Price**, providing all three columns allows the model to reconstruct Execution Price from the ratio of the other two. Even removing one leaky feature wouldn't fix the problem — the information persisted through a mathematical back-channel.

Feature	Baseline Importance	The Problem
Execution Price	51.5%	Bull-market time proxy — high price = recent = artificially profitable period. Non-causal.
Side_enc (BUY/SELL)	26.4%	Structural artifact — SELL fills close positions and show non-zero PnL; BUY fills always have PnL = 0. Tautological.
fg_value	18.0%	Legitimate signal — but drowned out by the two leaky features above.
Size USD	4.1%	Weak signal at fill level.
hour / day_of_week	<1%	Minimal importance at fill level.

6. Leakage Diagnosis & Temporal Validation

6.1 What is Data Leakage?

Data leakage occurs when information that would not be available at the moment of prediction is used during training. In trading, there are two types:

- **Look-ahead leakage:** The model is trained on future data, then tested on past data. The random split created this.
- **Feature leakage:** A feature is mechanically correlated with the target in a way that would not hold in real deployment — such as Execution Price encoding the time period.

6.2 Feature Removal Protocol

Features removed and the exact reason for each:

Execution Price — bull-run price trend makes it a time proxy; encodes market epoch rather than trade quality

Size Tokens — mathematically reconstructs Execution Price via the identity: Size Tokens = Size USD / Execution Price

Fee — post-execution computation, proportional to Size USD; indirect circular path back to trade size

Start Position — cumulative across prior trades; encodes the full history of preceding decisions (look-ahead risk)

log_size_usd — deterministic log transform of Size USD; carries zero new information for tree models

fg_cat — ordinal step-function encoding of fg_class; perfectly collinear with fg_value

The clean feature set was reduced to five variables: Size USD, Side_enc, fg_value, hour, and day_of_week.

6.3 The Temporal Split

A temporal split divides data by time rather than randomly. All fills before a cutoff date form the training set; all fills after form the test set. This mirrors real-world deployment: a model trained on history must predict the future.

Our split: Rows sorted chronologically. The first 80% (up to April 9, 2025) formed training. The last 20% (April 9 onward) formed the test set. The model could only learn from the past to predict the future — the only valid protocol for financial time-series.

6.4 Results After Removing Leakage

Model	CV AUC (training period)	Test AUC (future data)	Interpretation
Random Forest	0.658 ± 0.110	0.462	Below random — regime break detected
XGBoost	0.634 ± 0.086	0.470	Below random — same conclusion

Test AUC collapsed from 0.99 to 0.47 — below 0.5. This is not a model failure. An AUC below 0.5 means the model's learned signals inverted between the training and test periods. April 2025 coincided with a major macroeconomic shock (tariff announcements) and a sharp crypto market selloff. The patterns that predicted profitable trades in the 2024 bull market reversed entirely.

6.5 Cross-Validation Reveals the Regime Story

Time-series cross-validation with 5 expanding folds reveals when signal existed and when it did not:

Fold	Data Period	RF AUC	XGB AUC	What This Means
1	Earliest 20%	0.51	0.51	Near random — sparse early data, no learnable pattern
2	Up to 40%	0.55	0.55	Marginal signal beginning to emerge
3	Up to 60%	0.73	0.71	Stronger signal — bull market regime starting
4	Up to 80%	0.79	0.70	Peak signal — late 2024 bull run fully embedded in data
5	Up to 100%	0.73	0.71	Signal sustained but fading near test cutoff

The Side_enc feature still dominated the clean model at 36-57% importance — even without the leaky price features. This confirmed the fill-level structural artifact: SELL fills always have non-zero PnL; BUY fills always have PnL = 0. The model was predicting the fill type, not the outcome. Fixing this required moving from fill-level to trade-level data entirely.

7. Trade-Level Reconstruction

7.1 Why the Unit of Analysis Was Wrong

A fill is not a decision. The decision is the order. Modelling at the fill level is analogous to studying a football match by analysing each individual pass rather than the whole game — you miss the strategic intent. The correct unit of analysis is a complete round-trip trade: from position opening to position closing.

7.2 Aggregation by Order ID

We grouped all 211,224 fills by their Order ID, compressing them into 27,329 trade-level rows. Each output row represents one complete order and uses only information that was available at the time the order was first opened:

Output Column	Source Fills	Aggregation Method
open_price	BUY fills only	VWAP — Volume-Weighted Average Price across all BUY fills in the order
direction	All fills	LONG if BUY fills outnumber SELL fills, else SHORT
total_size_usd	All fills	Sum of all fill notional values
net_pnl	All fills	Sum of all Closed PnL values minus total fees paid
is_profitable	Derived	1 if net_pnl > 0, else 0 — the model's target
n_fills	Count	Number of sub-fills composing the order

7.3 Volume-Weighted Average Price (VWAP)

VWAP is the standard institutional measure of average execution quality. It weights each fill price by the dollar volume at that price, giving the true cost basis of the position.

Example: A trader buys ETH in three fills:

Fill 1: 100 tokens at $3,000 = $300,000 | Fill 2: 200 tokens at $3,010 = $602,000 | Fill 3: 50 tokens at $3,020 = $151,000

Simple average price: ($3,000 + $3,010 + $3,020) / 3 = $3,010

VWAP: ($300,000 + $602,000 + $151,000) / (100 + 200 + 50) = $3,008 — the true average cost per token

VWAP correctly reflects that most volume was bought at $3,010, pulling the average toward that price.

7.4 Impact of Restructuring

Metric	Fill-Level	Trade-Level	Why It Changed
Total rows	211,224	27,329	Multiple fills per order compressed into one row
Profitable rate	41.1%	24.7%	BUY fills (always PnL=0) artificially inflated the fill-level rate
Side_enc artifact	Dominant (36–57%)	Eliminated	Direction (LONG/SHORT) at trade level is genuinely meaningful
Unit of analysis	A sub-execution	A trading decision	Now correctly measures what the trader intended

The profitable rate dropping from 41.1% to 24.7% is a more honest representation. It reflects that roughly one in four trade decisions generated a net profit after fees — a challenging environment that demands a model with genuine signal.

8. Alpha Feature Engineering

In quantitative finance, alpha refers to excess return above what the market gives you — genuine edge. An alpha feature is any variable that carries predictive power not already embedded in the current price. The previous models described the trade itself but gave the model no information about what the market was doing before the decision was made. A human trader would never place a trade without looking at the chart. We needed to give the model the same context.

8.1 Price Momentum (1h, 4h, 24h)

Momentum is one of the most robustly documented phenomena in financial markets — assets that have been rising tend to continue rising over short to medium timeframes. This is called the momentum factor and has been validated across equities, bonds, commodities, and crypto for decades.

We computed the percentage price change for each coin over three lookback windows, using execution prices from prior fills of the same coin as an approximation of the price history:

- **mom_1h:** Percentage return over the last 60 minutes. Captures very short-term trend — has this coin's price been moving in the direction of this trade in the last hour?
- **mom_4h:** Percentage return over the last 4 hours. Intraday trend — is the broader trading session aligned with this position?
- **mom_24h:** Percentage return over the last 24 hours. Daily trend — what is the overall market doing today and is this trade in line with it?

8.2 Volatility (1h, 4h)

Volatility measures the magnitude of price fluctuations — the standard deviation of price observations over a time window. High volatility means prices are moving sharply; low volatility means prices are stable.

In options pricing (the Black-Scholes formula), volatility is the central input that determines option value. In directional trading, high volatility creates wider distributions of possible outcomes — bigger wins and bigger losses. A model that knows current volatility can calibrate its confidence in momentum signals, treating the same momentum reading differently in a calm vs turbulent environment.

- **vol_1h:** Standard deviation of execution prices in the last 60 minutes — immediate market turbulence.
- **vol_4h:** Standard deviation over 4 hours — broader intraday volatility context.

8.3 Relative Volume

Volume is one of the oldest confirmatory signals in technical analysis. The principle: a price move on unusually high volume is more significant than the same move on light volume. We defined

relative volume as the current fill's Size USD divided by the 7-day median fill size for the same coin. A value of 3.0 means this trade is three times the normal size — potentially indicating institutional order flow or a major market event.

8.4 Rolling Win Rate (Lagged)

This captures the trader's own recent performance momentum — the win rate over the preceding 20 trades. It is lagged by one trade to prevent look-ahead leakage. If a trader has won 17 of the last 20 trades, they may be in a productive strategy-market alignment. Markets go through periods where a given approach works consistently, then rotate. A high rolling win rate increases the probability that the current trade is in a productive regime.

8.5 Fee Ratio (Maker vs Taker Proxy)

In crypto exchanges, maker orders (limit orders that add liquidity) pay lower fees; taker orders (market orders that consume liquidity) pay higher fees. A high fee ratio signals urgency — the trader paid more to execute immediately. Fee ratio = total_fee / total_size_usd.

8.6 Complete Feature Set — 13 Variables

Feature	Category	Description
log_size	Position sizing	log(1 + total_size_usd) — compresses the extreme right skew of trade sizes into a usable range
direction_enc	Trade direction	1 = LONG (expecting price rise), 0 = SHORT (expecting price fall). Genuine signal at trade level.
fg_value	Market sentiment	Daily Fear & Greed Index score (0–100)
hour	Time context	Hour of trade execution (0–23) — crypto session liquidity patterns
day_of_week	Time context	0 (Monday) to 6 (Sunday) — weekend vs weekday trading dynamics
mom_1h	Price momentum	% price return of this coin in the prior 60 minutes
mom_4h	Price momentum	% price return of this coin in the prior 4 hours
mom_24h	Price momentum	% price return of this coin in the prior 24 hours
vol_1h	Volatility	Standard deviation of execution prices in the prior 60 minutes
vol_4h	Volatility	Standard deviation of execution prices in the prior 4 hours

Feature	Category	Description
fee_ratio	Execution quality	Fee as % of trade size — maker/taker urgency proxy
rolling_winrate	Trader performance	Win rate of the preceding 20 trades (lagged by 1 — no look-ahead)
n_fills	Order complexity	Number of sub-fills — complex orders may behave differently from simple ones

9. The Final Model — XGBoost vs Random Forest

9.1 Why These Two Algorithms?

Random Forest — Bagging Ensemble

Random Forest trains hundreds of decision trees simultaneously, each on a random bootstrap sample of rows and a random subset of features. This is called bagging — Bootstrap AGGregating. Each tree independently votes on the outcome; the final prediction is the majority vote or average. By averaging many uncorrelated trees, the ensemble is far more stable than any individual tree. For noisy financial data, this robustness is valuable — individual spurious patterns get diluted across the ensemble.

XGBoost — Sequential Gradient Boosting

XGBoost (Extreme Gradient Boosting) trains trees sequentially. Each new tree corrects the errors of the current ensemble, guided by the gradient of the loss function — the direction of steepest improvement. This boosting approach reduces bias (systematic underperformance) more aggressively than bagging. XGBoost adds L1 and L2 regularisation to the tree-building objective, explicitly penalising model complexity and preventing overfitting. It also handles missing values natively — critical because our momentum features have NaN values for early trades with insufficient price history.

9.2 Hyperparameter Choices for Financial Data

Financial data is notoriously noisy. Signals are weak and widespread — not deep and local. This requires conservative hyperparameter choices that prevent the model from memorising noise:

Parameter	Value	Why It Matters for Financial Data
max_depth (XGB)	4	Shallow trees. Financial signals don't require deep conditional logic — they are subtle patterns across many features.
max_depth (RF)	6	Slightly deeper because bagging already provides variance reduction through averaging.
min_child_weight	10	A leaf node must contain at least 10 samples. Prevents splitting on tiny subgroups that may be statistical accidents.
learning_rate	0.03	Slow, conservative updates. Compensated by 400 trees instead of 100. Generalises better than fast+shallow.
subsample	0.75	Each tree sees only 75% of training rows — adds randomness that prevents co-adaptation.
colsample_bytree	0.75	Each tree sees only 75% of features —

Parameter	Value	Why It Matters for Financial Data
		prevents any single feature from dominating every tree.
scale_pos_weight	~4.6x	With 24.7% profitable trades, up-weighting the minority class prevents the model from ignoring it entirely.

9.3 Class Imbalance — Why It Matters

With only 24.7% of trades being profitable, we have a class imbalance problem. A naive model that predicted not profitable for every single trade would achieve 75.3% accuracy — but be completely useless for the task. Three mechanisms were applied:

- **scale_pos_weight = ratio x 1.2:** The 1.2 multiplier over-corrects slightly to boost recall — catching more profitable trades at the cost of some precision.
- **class_weight='balanced' (Random Forest):** Automatically adjusts sample weights to equalise effective class frequencies during training.
- **Threshold tuning (post-training):** Moving the decision threshold from the default 0.5 to 0.65 further optimises the precision/recall trade-off for trading deployment.

9.4 Final Results

Metric	Random Forest	XGBoost	Interpretation
CV AUC (training)	0.861 ± 0.039	0.872 ± 0.043	Strong, consistent signal across all 5 temporal CV folds
Test AUC (future data)	0.877	0.910	Genuine out-of-sample performance on never-seen future trades
CV → Test Gap	-0.017	-0.037	Negative gap — model improves on the test set. Unusual and positive.
Accuracy	81.5%	81.0%	Correctly classifies 81% of trades overall
F1 Macro	0.762	0.759	Well-balanced performance across both profitable and unprofitable classes
Training Time	5.3 sec	0.7 sec	XGBoost is approximately 8x faster

XGBoost selected as the best model with Test AUC = 0.91. The negative CV-to-Test gap (the model actually performs better on the unseen test set than during cross-validation) suggests the most recent trading period is a particularly structured regime where price

momentum and sentiment signals are especially consistent and reliable.

9.5 Feature Importance in the Final Model

Feature	XGB Importance	Interpretation
mom_1h	Highest	Short-term price momentum is the dominant signal. What the price did in the last hour predicts whether this trade will be profitable.
mom_4h	High	Intraday trend confirmation — correlated with 1h but adds information about trend duration and sustainability.
rolling_winrate	Moderate	The trader's recent performance is genuinely predictive — when a strategy is aligned with the current regime, it runs in streaks.
fg_value	Moderate	Broader market sentiment adds signal beyond coin-specific price action alone.
vol_1h	Moderate	Current volatility conditions the interpretation of the momentum signal.
direction_enc	Lower	LONG vs SHORT is meaningful but secondary to what the price has been doing.
log_size	Low	Position size alone is a weak predictor of outcome quality.
n_fills	Lowest	Order complexity has minimal independent predictive power.

10. Threshold Analysis — From Model to Trading Tool

10.1 Why 0.5 Is Rarely the Right Threshold

A classification model outputs a probability between 0 and 1. The threshold is the probability above which the model predicts a positive class. By default this is 0.5, but different trading objectives demand different trade-offs.

Precision = Win Rate

Of all trades the model flags as profitable, what fraction actually were?

High precision = fewer false signals = less capital deployed on losing trades

Increase the threshold → higher precision, lower recall.

Recall = Capture Rate

Of all actually profitable trades that occurred, what fraction did the model identify?

High recall = fewer missed opportunities = more winners captured

Decrease the threshold → higher recall, lower precision.

10.2 Full Threshold Analysis

Threshold	Precision / Win Rate	Recall / Capture Rate	F1 Score	Trades Flagged
0.50	52.6%	84.5%	0.648	1,566 / 4,706
0.55	54.9%	81.1%	0.655	1,440 / 4,706
0.60	58.2%	77.6%	0.665	1,300 / 4,706
0.65 ★ Best F1	61.1%	74.3%	0.670	1,186 / 4,706
0.70	63.7%	68.5%	0.660	1,047 / 4,706
0.75	67.8%	61.5%	0.645	884 / 4,706
0.80	73.7%	52.1%	0.610	688 / 4,706
0.85	79.1%	40.8%	0.538	502 / 4,706
0.90	84.7%	26.1%	0.399	300 / 4,706

10.3 F1 Score — The Balancing Metric

The F1 Score is the harmonic mean of Precision and Recall: F1 = 2 × (Precision × Recall) / (Precision + Recall). It peaks where the two metrics are best balanced — at threshold 0.65 for this model (F1 = 0.670). The harmonic mean punishes extreme imbalances: a model with 100% precision and 0% recall has F1 = 0, correctly reflecting that it is operationally useless.

10.4 Recommendation by Use Case

Use Case	Threshold	Win Rate	Why
Balanced — maximise value	0.65	61.1%	Best F1. Takes 1,186 trades, wins 61% of them. Recommended default.
Conservative — capital preservation	0.80	73.7%	High conviction only. Misses 48% of profitable trades but very high precision.
Aggressive — maximise capture	0.50	52.6%	Catches 84.5% of winners but accepts more losses alongside them.
Research — highest confidence signals	0.90	84.7%	Only 300 signals — useful for studying the model's peak-confidence patterns.

Baseline comparison (no model): Randomly selecting trades yields a 24.7% win rate — the base rate of profitable trades in the test set.

With model at threshold 0.65: Win rate of 61.1% — a 2.5x improvement. The model provides genuine, actionable signal.

11. Key Findings & Future Improvements

11.1 What We Proved

1. **The unit of analysis was everything.** Moving from fill-level to trade-level data was the single most impactful change. No feature engineering or model tuning can fix fundamentally wrong data framing.
2. **Data leakage inflated AUC from 0.91 to 0.99.** Three subtle, compounding errors produced a publishable-looking but entirely false result. Explicit leakage diagnosis is non-negotiable in financial ML.
3. **Price momentum is the dominant signal.** The 1-hour and 4-hour momentum features became the top predictors after correct framing. This trader's profitability is tightly linked to short-term trend continuation.
4. **Sentiment matters, but non-linearly.** Fear & Greed carried meaningful signal in the final model — but only in combination with price context. Linear correlation (0.01) missed this entirely.
5. **Market regimes change abruptly.** The clean-feature model collapsed from 0.65 CV AUC to 0.47 test AUC because April 2025 broke sharply from 2024. No static model survives indefinitely without retraining.

11.2 What Would Improve the Model Further

- **External OHLCV data:** Proper candle data from an exchange API would make momentum and volatility features far more precise than the fill-price approximation used here.
- **Coin-specific relative strength:** The global Fear & Greed index treats all coins identically. A coin's return relative to Bitcoin measures whether it is outperforming or underperforming the broader market.
- **Funding rates:** In crypto perpetual futures, extreme positive funding rates signal overcrowded long positions — a leading indicator of reversal risk.
- **Liquidation heatmaps:** Clusters of leveraged positions at specific price levels create predictable momentum cascades when those levels are breached.
- **Walk-forward retraining:** Instead of training once on all historical data, retrain monthly on a rolling window to prevent regime staleness.
- **Alternative targets (MAE/MFE):** Predicting Maximum Adverse Excursion and Maximum Favorable Excursion enables risk/reward optimisation beyond simple binary profitability.

12. Glossary of Financial & Technical Terms

Term	Definition
Alpha	In finance: excess return above a market benchmark. In ML: a feature that carries genuine predictive signal beyond what is already priced in.
AUC (ROC-AUC)	Area Under the ROC Curve. Measures a model's ability to distinguish classes across all decision thresholds. 0.5 = random; 1.0 = perfect.
Bagging	Bootstrap Aggregating. Training multiple independent models on random data subsets and averaging predictions. Used by Random Forest.
Basis Points (bps)	1/100th of a percent. 10 bps = 0.1%. Standard unit for fees and small return differences in finance.
Beta	A coin's sensitivity to Bitcoin's price movements. Beta > 1 means the coin amplifies BTC moves; Beta < 1 means it dampens them.
Black-Scholes	The standard mathematical model for pricing options. Volatility is its central input — higher volatility = higher option price.
Boosting	Training models sequentially where each corrects the previous one's errors. Used by XGBoost. Reduces bias more aggressively than bagging.
Class Imbalance	When one target class (e.g., profitable trades at 24.7%) is far rarer than the other. Requires scale_pos_weight or class_weight to prevent the model from ignoring the minority class.
Closed PnL	Profit and Loss that has been realised by closing a position. Distinct from Unrealised (paper) PnL where the position is still open.
Correlation	A measure of linear relationship between two variables. Ranges from -1 (perfectly opposite) to +1 (perfectly aligned). Zero means no linear relationship — but non-linear relationships may still exist.
Cross-Validation	Technique to estimate how a model will generalise by training and testing on multiple subsets of data. TimeSeriesSplit preserves temporal ordering.
Data Leakage	When information unavailable at prediction time is used during training. Produces inflated performance metrics that fail in real deployment.
Decision Threshold	The probability cutoff above which a model predicts a positive class. Default 0.5 but often tuned to optimise precision vs recall.
Direction (LONG/SHORT)	LONG: position that profits when price rises. SHORT: position that profits when price falls. Distinct from fill-level BUY/SELL.
Drawdown	The peak-to-trough decline in portfolio value before recovery to a new peak. A key risk metric for trading strategies.
F1 Score	Harmonic mean of Precision and Recall: 2 x (P x R) / (P + R). Peaks

Term	Definition
	where the two are best balanced. Standard metric for imbalanced classification.
Fear & Greed Index	Daily composite sentiment indicator: 0 = Extreme Fear, 100 = Extreme Greed. Derived from volatility, volume, social data, dominance, and trends.
Fee Ratio	Fee / trade size. A proxy for maker vs taker order type — higher ratio signals urgency and willingness to pay for immediate execution.
Fill	A single sub-execution of a larger order. Large orders are typically filled across multiple price levels as liquidity becomes available.
FOMO	Fear Of Missing Out. The psychological driver behind buying into rising markets. Associated with Greed/Extreme Greed sentiment.
Funding Rate	Periodic payment between longs and shorts in perpetual futures. Extreme positive funding = overcrowded longs = elevated reversal risk.
Gradient Boosting	A boosting method that trains each tree to predict the gradient (error direction) of the previous ensemble's loss function.
Hyperparameter	A model setting chosen before training (e.g., max_depth, learning_rate). Not learned from data — tuned by the practitioner based on domain knowledge.
Kruskal-Wallis Test	Non-parametric statistical test for differences across multiple groups. Appropriate for non-normal financial data. Tests whether any group has a different distribution.
L1 / L2 Regularisation	Penalty terms added to the loss function to discourage overly complex models. L1 (Lasso) encourages sparsity; L2 (Ridge) shrinks all weights evenly.
Liquidity	How easily an asset can be bought or sold without moving its price. High liquidity = tight bid-ask spread, minimal slippage.
Look-Ahead Bias	Using future information to make past predictions during backtesting. The random split created this — future trades in training, past trades in testing.
MAE / MFE	Maximum Adverse Excursion (worst point reached against the trader before close) / Maximum Favorable Excursion (best point reached in the trader's favour).
Maker / Taker	Maker: limit order that adds liquidity to the book (lower fee). Taker: market order that removes liquidity (higher fee, signals urgency).
Market Regime	A sustained period characterised by specific market behaviour: bull market, bear market, ranging/sideways, high/low volatility.
Momentum	The tendency of assets trending in a direction to continue in that direction. One of the most documented factors in quantitative finance — validated across decades and asset classes.
Non-Stationarity	When statistical properties (mean, variance, correlations) of a time series change over time. Almost all financial data is non-stationary, making standard cross-validation invalid.

Term	Definition
OHLCV	Open, High, Low, Close, Volume — the standard candlestick data format for price history.
On-Chain	Data recorded directly on the blockchain: transaction history, wallet balances, smart contract interactions. Transparent and immutable.
Open Interest	The total number of outstanding derivatives contracts not yet settled. High open interest = large leveraged positions — potential fuel for liquidation cascades.
Order	A trader's instruction to buy or sell an asset at specific terms. May be filled as one or many sub-executions.
Overfitting	When a model learns patterns specific to training data that do not generalise to new data. Symptom: very high training performance, significantly lower test performance.
Perpetual Futures	Derivative products that track spot prices with no expiry date. The most popular trading instrument in crypto. Funded by periodic payments (funding rate) between longs and shorts.
PnL	Profit and Loss. The financial result of a trading decision. Net PnL = Gross PnL minus all fees.
Precision	Of all trades predicted positive, what fraction were truly positive. Called Win Rate in a trading context.
Recall	Of all truly positive cases, what fraction were correctly identified. Called Capture Rate or Hit Rate in trading.
Regime Change	A structural shift in market behaviour that invalidates patterns learned in the previous period. The April 2025 selloff was one such change.
Relative Strength	A coin's performance relative to a benchmark (usually Bitcoin). Positive relative strength means outperforming BTC.
Relative Volume	Current volume divided by average historical volume. Values above 1 indicate unusual trading activity.
ROC Curve	Receiver Operating Characteristic. Plots True Positive Rate vs False Positive Rate at every threshold. AUC summarises the entire curve in one number.
Round-Trip Trade	A complete trading cycle: opening a position and subsequently closing it. The natural unit for measuring trading decisions, not fills.
Scale Pos Weight	XGBoost parameter that up-weights the minority class. Set to (negative count / positive count) to compensate for class imbalance.
Sharpe Ratio	Risk-adjusted return: mean return / standard deviation of returns. Higher = better return per unit of risk taken.
Short	A position that profits when the price falls. In crypto, executed via borrowed tokens or futures contracts.
Skewness	Asymmetry in a probability distribution. Positive skew means a long right tail. Financial returns and trade sizes are typically positively skewed.
Slippage	The difference between the expected execution price and actual

Term	Definition
	price. Higher in fast-moving or illiquid markets.
Temporal Split	Dividing time-series data by time, not randomly. Training on the past, testing on the future — the only valid approach for financial data.
TimeSeriesSplit	Scikit-learn's cross-validation for temporal data. Each fold trains on all preceding data and validates on the immediately following window.
Volatility	The statistical measure of price fluctuation, typically the standard deviation of returns over a window. High volatility = greater uncertainty and risk.
VWAP	Volume-Weighted Average Price. The average price paid, weighted by volume executed at each price level. The true cost basis of an institutional position.
Walk-Forward Validation	Rigorous backtesting where the model is retrained at regular intervals using only historically available data — simulating real deployment.
Win Rate	The percentage of trades that were profitable. Equivalent to Precision in the ML context when predicting trade outcomes.
XGBoost	Extreme Gradient Boosting. A fast, regularised, sequential tree-building library. The industry benchmark algorithm for tabular data machine learning tasks.

End of Report